MILITARY SPECIFICATION

DELAY ELEMENT, CARTRIDGE: M60 ASSEMBLY

1. SCOPE

1.1 This specification covers Delay Element, Cartridge: M60, Assembly, for use in cartridges for propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-B-550	- Barium Chromate
MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-P-46994	- Boron, Potassium Nitrate
MIL-B-51092	- Boron, Amorphous, Powder

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions

DRAWINGS

Munitions Command

D10536004	- Delay, Element, Cartridge, M60, Assembly
C8826728	- Ammunition Container, Metal, M21 for Delay Element M5, M8, M13, M14, M15, M16, M17 or M60 (see 5.1)

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2.1 (Cont'd)

DRAWINGS

C8826729	- Box, Packing, Ammunition for Delay Element, M5, M8, M13, M14, M15, M16, M17 or M60 (see 5.1)
IEL 10536004	- Inspection Equipment List for Delay Element, Cartridge, M60, Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N. Y. 10018.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Protective finish.-Protective finish shall be in accordance with the requirements specified on the applicable drawing and referenced specification and shall apply to all surfaces regardless of configuration.

3.3 Vibration.-The delay element assembly shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer or delay charge or derangement of metal parts.

3.4 Functional.

3.4.1 The delay element shall not fail to fire.

3.4.2 Ballistics.-The delay charge shall consist of Barium Chromate as specified in MIL-B-550, and Boron as specified in MIL-B-51092, in the proportion and quantity fixed by the contractor to produce the delay time required in Table I (see 6.4).

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Table I

Temperature	Delay Time, Seconds (see 6.2)
-65° to -70°F	0.87 max.
65° to 75°F	0.72 ± 0.10*
200° to 205°F	0.54 min.

* For information only.

3.5 Workmanship.- Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following:

3.5.1 Metal defects.- The part shall be free from cracks, splits, cold shuts, inclusions and porosity and other metal defects.

3.5.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facility suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall

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consist of 23 anodized units for protective finish tests, 30 assembled units for ballistics and 10 sets of unassembled components for visual and dimensional examination.

The sample shall be manufactured using the same material, equipment, processes, and procedures as used in regular production. All parts and materials including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and tests.-After examination and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.-The sample for ballistics tests shall be temperature conditioned and tested to determine compliance with Table I. The sample for firing shall consist of the following:

Table II

Temperature	Not Vibrated	Vibrated
-65° to -70°F	6	4
65° to 75°F	6	4
200°to 205°F	6	4

4.2.1.3 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submittal of product.-The contractor shall submit the product in accordance with MIL-STD-105.

4.3.1.2 Size of lot.-A lot shall consist of an identifiable group of delay element assemblies consisting of:

a. Components produced by a homogeneous process.
b. Delay charge from not more than one lot.
c. Primers from not more than one lot.

4.3.2 Examination.-One-hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be

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performed on a defect basis in accordance with the classification of defects, using the sampling plan and associated criteria in Tables III and IV. For an individual defect (single characteristic), the criteria in Table III shall be used. All non-conforming material shall be rejected. Four power magnification (4X) may be used in performing visual examination for such defects as burr and foreign matter. The roughness comparison specimens prescribed in ASA-B46.1 shall be used as a basis of comparison for surface roughness determination.

Table III

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

Table IV

	Total Allowable Defects Per Sample Size			
Sample Size for Major Defects Number of Defects in Class	110	150	225	300
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14
Sample Size for Minor Defects	**110**	**150**	**225**	**300**
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

4.3.2.1 Body, Delay, Dwg. C8594081, covering a detail of Drawing D10536004.

Defect	Method of Inspection
CRITICAL	
1. Diameter of flash hole	Gage
2. Metal defective	Visual

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	Defect	Method of Inspection
MAJOR		
101.	Pitch diameter of external thread, min.	Gage
102.	Major diameter of external thread, min.	Gage
103.	Concentricity of body with pitch diameter of thread	Gage
104.	Diameter of thread relief	Gage
105.	Diameter of step at thread relief	Gage
106.	Diameter of cavity, max.	Gage
107.	Diameter of body, min.	Gage
108.	Depth of cavity, max.	Gage
109.	Length from cavity face to body diameter	Gage
110.	Thread damaged or not full	Visual
111.	Protective finish inadequate or defective	Visual
112.	Burr	Visual
113.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of thread, max.	Gage
202.	Diameter of cavity, min.	Gage
203.	Diameter of body, max.	Gage
204.	Diameter of wrench holes	Gage
205.	Depth of cavity, min.	Gage
206.	Depth of wrench holes	Gage
207.	Total length	Gage
208.	Length of thread pilot	Gage
209.	Length of thread relief	Gage
210.	Length of step at thread relief	Gage
211.	Location of wrench holes	Gage
212.	Radius or chamfer missing or incorrect	Visual
213.	Surface finish improper	Visual

4.3.2.2 Gasket, Aluminum, Drawing B8593082, covering a detail of Dwg. D10536004.

CRITICAL		
	None defined	
MAJOR		
101.	Diameter outside, min.	Gage
102.	Diameter inside, max.	Gage

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Defect	Method of Inspection
103. Metal defective	Visual
104. Burr	Visual
105. Foreign matter	Visual

MINOR

201. Diameter, outside, max.	Gage
202. Diameter, inside, min.	Gage
203. Thickness	Gage
204. Surface finish improper	Visual

4.3.2.3 Primer, Retainer, Dwg. C8594083, covering a detail of Dwg. D10536004.

CRITICAL

1. Diameter of flash hole	Gage
2. Depth of primer pocket	Gage
3. Metal defective	Visual

MAJOR

101. Pitch diameter of internal thread, max.	Gage
102. Minor diameter of internal thread, max.	Gage
103. Concentricity of head with body	Gage
104. Concentricity of pitch diameter of thread with body	Gage
105. Concentricity of primer pocket with body	Gage
106. Concentricity of recess around primer pocket with primer pocket	Gage
107. Diameter of body, min.	Gage
108. Diameter of head	Gage
109. Diameter of primer pocket	Gage
110. Diameter at base of angled recess	Gage
111. Diameter of recess around primer pocket	Gage
112. Depth of cavity	Gage
113. Depth of recess around primer pocket	Gage
114. Depth of chamfer at primer pocket	Gage
115. Length of effective thread, min.	Gage
116. Location of datum diameter	Gage
117. Thread damaged or not full	Visual
118. Protective finish inadequate or defective	Visual
119. Surface finish of primer pocket improper	Visual
120. Burr	Visual

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Defect	Method of Inspection
121. Foreign matter	Visual
MINOR	
201. Pitch diameter of internal thread, min.	Gage
202. Diameter of body, max.	Gage
203. Total length	Gage
204. Length of wrench flats	Gage
205. Width across wrench flats	Gage
206. Location of wrench flats	Gage
207. Radius or chamfer missing or incorrect	Visual
208. Surface finish improper, except as otherwise classified	Visual

4.3.2.4 Primer, Percussion, M42, Drawing B8799925, covering a detail of Drawing D10536004.

CRITICAL	
1. Improper assembly	Visual
2. Any part missing	Visual
3. Primer pellet broken, cracked or dislodged	Visual
4. Metal defective	Visual
5. Primer foil discolored	Visual
6. Mixed primers	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.3.2.5 Assembly, Drawing D10536004.

CRITICAL	
1. Improper assembly	X-Ray 1/ & Visual
2. Any part missing	X-Ray 1/ & Visual
3. Primer not seated to full depth of primer pocket	Visual
4. Marking missing, incorrect or illegible	Visual

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Defect		Method of Inspection
MAJOR		
101.	Concentricity of flange with body	Gage
102.	Junction of delay body, gasket and primer retainer not properly sealed	Visual
103.	Junction of top, edge of primer pocket and crimp not properly sealed for 360°	Visual
104.	Burr	Visual
105.	Foreign matter	Visual
MINOR		
201.	Diameter of body, max.	Gage
202.	Total length, max.	Gage

1/ The delay element shall be x-rayed with the element in a position horizontal to the rays of the x-ray machine. The negatives properly identified shall be forwarded to the procuring activity.

4.3.2.6 Ammunition Container, Metal, M21, Drawing C8826728.

CRITICAL

None defined

MAJOR

None defined

MINOR

201.	Contents loose	Manual & Visual
202.	Quantity of content incorrect	Visual
203.	Separator or filler missing	Visual
204.	Heat seal incomplete, loose or distorted	Visual
205.	Container cut or damaged	Visual
206.	Marking missing, incorrect or illegible	Visual

4.3.2.7 Box, Packing, Ammunition for Delay Element, C8826729.

CRITICAL

None defined

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4.3.2.7 (Cont'd)

Defect	Method of Inspection
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual & Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of cartons incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Protective finish.-The number of units for protective finish tests shall be in accordance with the applicable specification referenced on the drawing. Failure of any sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.3.3.2 Vibration.-Twelve of the 30 assembled x-rayed delay elements of the initial production sample shall be subjected to vibration as directed in 3.3. After the vibration test, the 12 vibrated delay elements shall be x-rayed as directed in 1/ of 4.3.2.5. Evidence of any disintegration of the primer or delay charge or derangement of any component shall be cause for rejection of the lot. When it becomes necessary to make a change in the method of changing the delay element or in the lot of primers after the initiation of production, 12 delay elements from the first production lot thereafter shall be tested as provided herein.

4.3.3.3 Functional.-The following test shall be performed by the Government.

4.3.3.1 Ballistics.-The 30 test delay elements, including the 12 vibrated delay elements of the initial production sample (or subsequent sample of 12 delay elements when a change has been made in the method of charging the delay element or lot of primers) shall be temperature conditioned with the balance of the ballistic sample in accordance with Table II and 4.4.4. Failure of any delay element to comply with the requirement of 3.4, or failure of a sample to meet the criteria of the variables sampling plan, shall be cause for rejection of the lot.

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4.3.4 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL 10536004.

4.4 Test methods and procedures.

4.4.1 Protective finish.-Protective finish tests shall be in accordance with the specifications listed on the applicable drawing and referenced specification.

4.4.2 Vibration.-The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.3 Functional.

4.4.3.1 Ballistics.-The 30 assembled delay elements of each production lot shall be grouped and temperature conditioned in accordance with Table II and 4.4.3.1.1.

4.4.3.1.1 Temperature conditioning.-The vent holes of the delay elements shall be sealed against environmental atmosphere of the conditioning chamber. The temperature of the delay charge shall be considered as the conditioning temperature and will be determined by means of a thermocouple attached to the center of the delay charge of a specimen assembled delay element. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the specimen delay element to the required temperature. Upon completion of the temperature conditioning, the vent hole seal shall be removed and the delay elements tested within 5 minutes after removal from the conditioning chamber. If this time limit is exceeded, the sample shall be returned to the conditioning chamber and reconditioned as indicated herein. The total conditioning time, including reconditioning at 200° to 205°F shall not exceed 18 hours.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Level A.-Packing and marking shall be in accordance with drawings C8826728 and C8826729, except that the marking shall refer to Delay Element, Cartridge XM60, Assembly.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 Delay time is defined as the time lapse between the impact of the firing pin on the primer and evidence of gas flow from the vent hole of the delay element.

6.3 Advisory output material (Formulation).-Approximately 23.7 percent boron, 70.7 percent potassium nitrate and 5.6 percent binder. Advisory individual delay element output material weight: approximately 1.5 grains.

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6.4 Advisory delay charge composition (formulation).-Approximately 95.0 percent barium chromate and 5.0 percent boron. Advisory individual delay element charge weight: approximately 14 grains.

6.5 Loaded assembly hazard information.-

Explosive Hazard Classification: Class 3
Storage Compatibility: Group B, E, or N
Department of Transportation DOT Hazard : Class C
DOT Marking (Small Arms Ammunition) (Keep Fire Away): Class C

6.6 Hazard notice.-The delay element cartridge described herein and certain of its components are flammable and/or explosive; consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

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